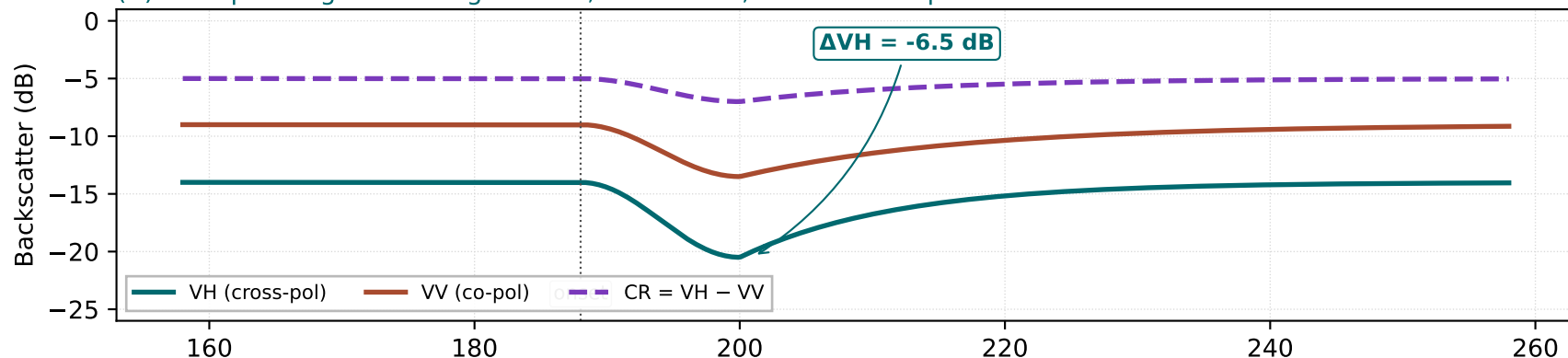
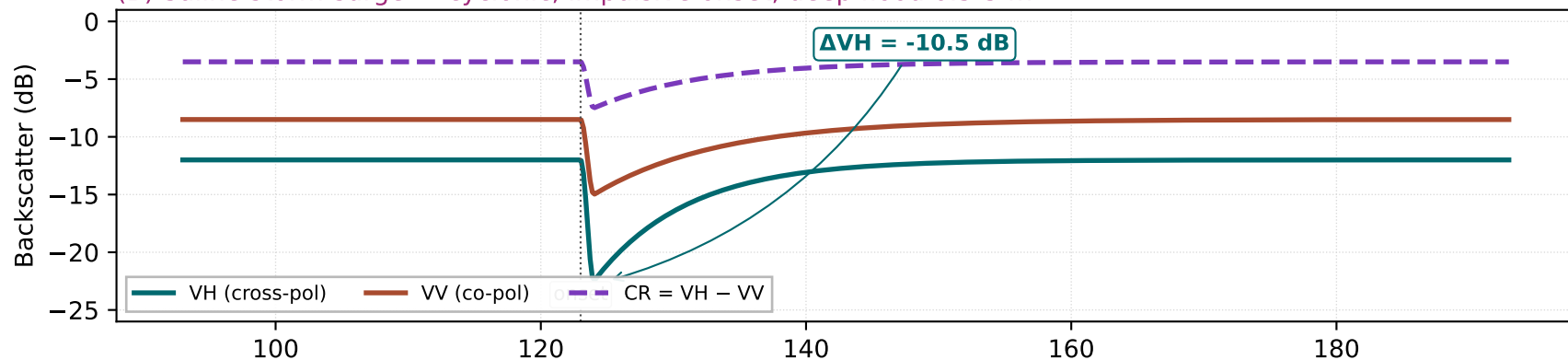


Figure S2. Canonical Sentinel-1 dual-polarisation backscatter signatures for the three inundation mechanisms. Differentiable feature space exploited by the Module 02 random-forest classifier.

(A) Transplanting flood — agronomic, freshwater, controlled depth ~5-15 cm



(B) Saline storm-surge — cyclonic, impulsive onset, deep flood 0.5-3 m



(C) Freshwater rainfall (Bulbul-class) — post-monsoon, shallow ponded water

